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Benchmarking Formula 1 results using a normal model

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ABSTRACT

There is enduring interest in disentangling the effects of skill and luck in sport. A key issue in Formula 1 is distinguishing between car-level and driver-level effects. Four elite teams currently dominate Formula 1 and have won every major race for the last four years. In this paper, we use univariate and bivariate normal models to quantify reasonable performance expectations at both driver and team levels, distinguishing between elite and non-elite teams. We illustrate our approach with an application to the last fully completed 2025 season.

KEYWORDS

Bivariate normal distribution; Formula 1; non-independence; sports

JEL CLASSIFICATION

C10; L83; Z20

1. Introduction

Key questions in sports analysis concern what constitutes reasonable performance expectations (Fry, Serbera, and Wilson 2021) and how to distinguish the effects of luck and skill (Scarf, Khare, and Alotaibi 2022). Though often under-explored, Formula 1 and motorsport provide a rich source of economic problems (Rockerbie and Easton 2022), particularly given Formula 1's rapid recent expansion.

Analysing Formula 1 is complicated by the need to disentangle car-level and driver-level effects (Fry, Brighton, and Fanzon 2024). Teams also pursue objectives beyond winning like improving on previous seasons, outperforming close rivals, securing higher prize-money, and strengthening longer-term competitive standing (Mourão, Macedo, and Oliveira 2025).

At the time of writing, Formula 1 consists of ten teams. Four elite teams (Red Bull, Mercedes, Ferrari, McLaren) dominate the sport and had won every major race over the past four seasons. Competition within these confines remains intense. This paper investigates reasonable performance expectations under these conditions, explicitly accounting for the distinction between elite and non-elite teams. Furthermore, within a team, driver finishing positions are negatively correlated, violating the independence assumption common to most sports' models (Scarf, Parama, and McHale 2019). We

develop a framework based on univariate and bivariate normal distributions that addresses both the elite/non-elite distinction and within-team correlations. This enables a realistic evaluation of performance, and captures strategic and competitive aspects of the sport. We obtain analytical expressions for winning probabilities and supplement these with Monte Carlo simulation analyses. These simulations establish performance benchmark expectations for elite and non-elite drivers and teams that can be compared with actual results.

Our analysis reveals that elite drivers Norris, Piastri and Verstappen exceeded expectations. However, several other elite-team drivers seemingly under-perform. Many non-elite drivers also exceeded expectations, highlighting the depth of talent in the sport. It is important to emphasize that non-elite teams also include exceptional drivers, notably Fernando Alonso (widely regarded as one of the best drivers but arguably constrained by a relatively modest Aston Martin car). Fry, Brighton, and Fanzon (2024) provide statistical evidence of Alonso outperforming his car. At the team level, McLaren exceeded expectations, whilst other elite teams Red Bull, Ferrari and Mercedes under-performed. Results obtained thus cast the sport in a new light and offer a more nuanced understanding of performance across teams and drivers.

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This paper is organized as follows. [Section II](#) presents a univariate model for individual driver performance. [Section III](#) develops a bivariate model for team performance. [Section IV](#) gives an empirical application to the last fully completed 2025 Formula 1 season. [Section V](#) concludes. Additional benchmarking considerations are discussed in an [Appendix](#) at the end of this paper.

II. Univariate modelling of driver performance

This section outlines a univariate model to describe the performance of individual drivers. There are four elite teams and six non-elite teams. Each team has two drivers. We assume that successive races are independent. Let the final finishing position in a race be approximated by a normal distribution. Whilst a strong assumption, this is consistent with the use of regression-based models in applications (Fry, Brighton, and Fanzon 2024). For elite-team drivers, set

$$E[\text{Elite driver Finishing Position}] = E[U\{1, 8\}] = 4.5, \quad (1)$$

where $U\{1, 8\}$ denotes a discrete uniform distribution over the integers 1–8. Similarly, for non-elite drivers set

$$\begin{aligned} E[\text{Non – elite driver Finishing Position}] \\ = E[U\{9, 20\}] = 14.5, \end{aligned} \quad (2)$$

where $U\{9, 29\}$ denotes a discrete uniform distribution over the integers 9–20. Elite drivers thus typically occupy top positions, with non-elite drivers finishing lower. The finishing rank of an elite-team driver is modeled as $N(4.5, \sigma_E^2)$. Similarly, the rank of a non-elite team driver is modeled as $N(14.5, \sigma_N^2)$. Thus, the probability an elite-team driver wins is

$$\begin{aligned} \Pr(\text{Win}) &= \Pr(\text{Rank} \leq 1.5) = \Phi\left(\frac{1.5 - 4.5}{\sigma_E}\right) \\ &= \Phi\left(-\frac{3}{\sigma_E}\right), \end{aligned} \quad (3)$$

where $\Phi(\cdot)$ denotes the $N(0, 1)$ Cumulative Distribution Function (CDF). If one of the eight elite-team drivers is guaranteed to win (consistent with the last four completed seasons)

$$8\Phi\left(-\frac{3}{\sigma_E}\right) = 1; \sigma_E = -\frac{3}{\Phi^{-1}\left(\frac{1}{8}\right)} = 2.607903. \quad (4)$$

Similarly, the probability that a non-elite team driver finishes in the top 9 positions is

$$\Pr(\text{Rank} \leq 9.5) = \Phi\left(\frac{9.5 - 14.5}{\sigma_N}\right) = \Phi\left(-\frac{5}{\sigma_N}\right). \quad (5)$$

Since there are only 8 elite-team drivers, at least one of the 12 non-elite drivers is guaranteed to finish in the top 9. Hence

$$12\Phi\left(-\frac{5}{\sigma_N}\right) = 1; \sigma_N = -\frac{5}{\Phi^{-1}\left(\frac{1}{12}\right)} = 3.615344. \quad (6)$$

Equation (6) indicates caution is needed when comparing drivers from different teams as the performance of non-elite drivers is inherently more variable with $\sigma_N > \sigma_E$.

Continuing, [Table 1](#) presents analytical expressions for the probabilities of finishing in the points-scoring positions shown in [Table 2](#). These quantities may be of particular interest to teams beyond winning (Mourão, Macedo, and Oliveira 2025). These results notwithstanding Monte Carlo simulation simplifies the analysis. For example, results for elite-team drivers can be simulated using

- (i) Simulate a finishing position from $N(4.5, \sigma_E^2)$.
- (ii) Round to the nearest integer between 1 and 20.
- (iii) Allocate points according to [Table 2](#).
- (iv) Repeat Steps (i)–(iii) for the remaining races in a season.

III. Bivariate modelling of team performance

This section develops a bivariate model for joint performance of two drivers within the same team. This explicitly accounts for negative correlations in teammates' finishing positions. The joint performance of the two drivers in the same elite team is modelled by a bivariate normal distribution with the same marginal distributions as in [Section II](#):

Table 1. Analytical expressions for race outcome probabilities.

Probability	Elite-Team Driver	Non-Elite Team Driver
Pr(Win)	$\Phi\left(-\frac{3}{\sigma_E}\right)$	$\Phi\left(-\frac{13}{\sigma_N}\right)$
Pr(Second)	$\Phi\left(-\frac{2}{\sigma_E}\right) - \Phi\left(-\frac{3}{\sigma_E}\right)$	$\Phi\left(-\frac{12}{\sigma_N}\right) - \Phi\left(-\frac{13}{\sigma_N}\right)$
Pr(Third)	$\Phi\left(-\frac{1}{\sigma_E}\right) - \Phi\left(-\frac{2}{\sigma_E}\right)$	$\Phi\left(-\frac{11}{\sigma_N}\right) - \Phi\left(-\frac{12}{\sigma_N}\right)$
Pr(Fourth)	$\frac{1}{2} - \Phi\left(-\frac{1}{\sigma_E}\right)$	$\Phi\left(-\frac{10}{\sigma_N}\right) - \Phi\left(-\frac{11}{\sigma_N}\right)$
Pr(Fifth)	$\Phi\left(\frac{1}{\sigma_E}\right) - \frac{1}{2}$	$\Phi\left(-\frac{9}{\sigma_N}\right) - \Phi\left(-\frac{10}{\sigma_N}\right)$
Pr(Sixth)	$\Phi\left(\frac{2}{\sigma_E}\right) - \Phi\left(\frac{1}{\sigma_E}\right)$	$\Phi\left(-\frac{8}{\sigma_N}\right) - \Phi\left(-\frac{9}{\sigma_N}\right)$
Pr(Seventh)	$\Phi\left(\frac{3}{\sigma_E}\right) - \Phi\left(\frac{2}{\sigma_E}\right)$	$\Phi\left(-\frac{7}{\sigma_N}\right) - \Phi\left(-\frac{8}{\sigma_N}\right)$
Pr(Eighth)	$\Phi\left(\frac{4}{\sigma_E}\right) - \Phi\left(\frac{3}{\sigma_E}\right)$	$\Phi\left(-\frac{6}{\sigma_N}\right) - \Phi\left(-\frac{7}{\sigma_N}\right)$
Pr(Ninth)	$\Phi\left(\frac{5}{\sigma_E}\right) - \Phi\left(\frac{4}{\sigma_E}\right)$	$\Phi\left(-\frac{5}{\sigma_N}\right) - \Phi\left(-\frac{6}{\sigma_N}\right)$
Pr(Tenth)	$\Phi\left(\frac{6}{\sigma_E}\right) - \Phi\left(\frac{5}{\sigma_E}\right)$	$\Phi\left(-\frac{4}{\sigma_N}\right) - \Phi\left(-\frac{5}{\sigma_N}\right)$
Pr(Podiumfinish)	$\Phi\left(-\frac{1}{\sigma_E}\right)$	$\Phi\left(-\frac{11}{\sigma_N}\right)$
Pr(Top8finish)	$\Phi\left(\frac{4}{\sigma_E}\right)$	$\Phi\left(-\frac{6}{\sigma_N}\right)$
Pr(Top10finish)	$\Phi\left(\frac{6}{\sigma_E}\right)$	$\Phi\left(-\frac{4}{\sigma_N}\right)$

Table 2. Points awarded to formula 1 drivers for full races and sprint races in the 2025 season.

Full Race		Sprint Race	
Position	Points	Position	Points
1	25	1	8
2	18	2	7
3	15	3	6
4	12	4	5
5	10	5	4
6	8	6	3
7	6	7	2
8	4	8	1
9	2		
10	1		

$$\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 4.5 \\ 4.5 \end{pmatrix}, \begin{pmatrix} \sigma_E^2 & \sigma_{EE} \\ \sigma_{EE} & \sigma_E^2 \end{pmatrix}\right), \quad (7)$$

where r_i denotes the rank of Driver i . The covariance σ_{EE} in Equation (7) can be calculated as follows. From the linear transformation property of the normal distribution the sum of the ranks satisfies $r_1 + r_2 \sim N(9, 2\sigma_E^2 + 2\sigma_{EE})$. Further, $\Pr(r_1 + r_2 \leq 3) = 0$, giving

$$\Pr(r_1 + r_2 \leq 3) = \Phi\left(\frac{3 - 9}{\sqrt{2\sigma_E^2 + 2\sigma_{EE}}}\right) = 0.$$

Tabulated values in Neave (1978) give $\Phi(-4.9) = 0$. Thus,

$$\begin{aligned} \frac{3 - 9}{\sqrt{2\sigma_E^2 + 2\sigma_{EE}}} &= -4.9; \quad \sigma_{EE} = \frac{36}{48.02} - \sigma_E^2 \\ &= -6.051472. \end{aligned} \quad (8)$$

For a non-elite team Equation (7) is replaced by

$$\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 14.5 \\ 14.5 \end{pmatrix}, \begin{pmatrix} \sigma_N^2 & \sigma_{NN} \\ \sigma_{NN} & \sigma_N^2 \end{pmatrix}\right). \quad (9)$$

The covariance in Equation (9) can be calculated as follows. From the linear transformation property of the normal distribution we have that $r_1 + r_2 \sim N(29, 2\sigma_N^2 + 2\sigma_{NN})$. Further, we have that $\Pr(r_1 + r_2 \leq 39) = 1$. Tabulated values in Neave (1978) give $\Phi(4.9) = 1$, so

$$\begin{aligned} \frac{39 - 29}{\sqrt{2\sigma_N^2 + 2\sigma_{NN}}} &= 4.9; \quad \sigma_{NN} = \frac{100}{48.02} - \sigma_N^2 \\ &= -10.98825. \end{aligned} \quad (10)$$

Analysis can be simplified by Monte Carlo simulation. For example, results for negatively correlated drivers in the same elite-team can be simulated using

- (1) Simulate finishing positions from $N\left(\begin{pmatrix} 4.5 \\ 4.5 \end{pmatrix}, \begin{pmatrix} \sigma_E^2 & \sigma_{EE} \\ \sigma_{EE} & \sigma_E^2 \end{pmatrix}\right)$.
- (2) Round to the nearest integers between 1 and 20.
- (3) Allocate points according to Table 2.
- (4) Repeat Steps 1–3 for the remaining races in a season.

Table 3. Monte Carlo simulation of total season points under the 2025 formula 1 scoring system (Table 2), based on 1,000,000 simulations.

Category	Mean points	95% CI
Elite driver	315.456	(253–381)
Elite team	630.91	(594–669)
Non-elite driver	10.636	(1–29)
Non-elite team	21.305	(5–44)

Table 4. Total points scored by drivers in the 2025 season (elite-team drivers are highlighted). Symbols indicate performance relative to Monte Carlo CIs: ↑ = above, → = meeting, ↓ = below.

Driver	Team	Points	Performance
Lando Norris	McLaren	423	↑
Oscar Piastri	McLaren	410	↑'
George Russell	Mercedes	319	→
Kimi Antonelli	Mercedes	150	↓
Max Verstappen	Red Bull	421	↑
Yuki Tsunoda	Red Bull	33	↓
Charles Leclerc	Ferrari	242	↓
Lewis Hamilton	Ferrari	156	↓
Alexander Albon	Williams	73	↑
Carlos Sainz Jr	Williams	64	↑
Isack Hadjar	Racing Bulls	51	↑
Liam Lawson	Racing Bulls	38	↑
Fernando Alonso	Aston Martin	56	↑
Lance Stroll	Aston Martin	33	→
Oliver Bearman	Haas	41	↑
Esteban Ocon	Haas	38	↑
Nico Hülkenberg	Sauber	51	↑
Gabriel Bortoleto	Sauber	19	→
Pierre Gasly	Alpine	22	→
Franco Colapinto	Alpine	0	↓

Table 5. Total points scored by teams in the 2025 season (elite teams are highlighted). Symbols indicate performance relative to Monte Carlo CIs: ↑ = above, → = meeting, ↓ = below.

Team	Points	Performance
McLaren	833	↑
Mercedes	469	↓
Red Bull	454	↓
Ferrari	398	↓
Williams	137	↑
Racing Bulls	92	↑
Aston Martin	89	↑
Haas	79	↑
Sauber	70	↑
Alpine	22	→

IV. Empirical applications

This section compares results in Section II–3 with the last fully completed 2025 season. Table 3 gives the expected value and 95% confidence intervals (CIs) for season-long points totals, across elite and non-elite drivers and teams, calculated by Monte Carlo simulation. These CIs provide a benchmark for reasonable performance expectations when compared with actual results.

Results for individual drivers are reported in Table 4. Several elite drivers exceed expectations (Norris, Piastri, and Verstappen). Some elite drivers may be under-performing (Antonelli, Hamilton, Leclerc and Tsunoda). Remarkably, most non-elite drivers exceed expectations, highlighting the talent and depth of competition in Formula 1. Fernando Alonso is a particularly highly rated driver (Fry, Brighton, and Fanzon 2024) outperforming the benchmark. Colapinto is the only non-elite driver potentially under-performing. Results in Table 5 provide further insights into team performance. McLaren, exceed expectations after an outstanding season. In contrast, the remaining elite teams seem to under-perform. Most non-elite teams exceed expectations. Even Alpine perform in line with reasonable performance expectations.

V. Conclusions

This paper quantifies reasonable performance expectations in sports (Fry, Serbera, and Wilson 2021), linked to related questions of competitive balance (Plumley, Ramchandani, and Wilson 2019) and performance efficiency (Rossi et al. 2019). We also contribute to a wider literature disentangling driver and car-level effects in Formula 1 (Bell et al. 2016; Fry, Brighton, and Fanzon 2024; Van Kesteren and Bergkamp 2023).

This paper develops univariate and bivariate models to assess reasonable performance expectations for drivers and teams. By explicitly incorporating the elite/non-elite distinction and within-team correlations, the models enable a realistic evaluation of performance. We derive analytical results for winning probabilities, and conduct Monte Carlo simulations. These simulations provide benchmarks for drivers and teams, enabling a rigorous evaluation of performance in the last fully completed 2025 season. However, Formula 1 remains highly competitive and random, so race results must be interpreted cautiously. Sometimes performance expectations may need to be revised downward (see the Appendix).

Empirical results obtained contribute to the wider literature (Bell et al. 2016; Fry, Brighton, and Fanzon 2024; Van Kesteren and Bergkamp 2023). At the driver level, Norris, Piastri, and Verstappen exceeded expectations. Several elite drivers under-perform

relative to model predictions. At the team level, McLaren exceeded expectations, whilst other elite teams Mercedes, Red Bull and Ferrari underperform. Remarkably, most non-elite drivers exceed expectations, with Colapinto the only exception. This emphasizes the depth of talent in the sport. Overall results offer a more nuanced and quantitative understanding of outcomes in Formula 1. Potential applications include driver selection and contract negotiations. However, there remains an enduring need to interpret the numbers carefully. The [Appendix](#) discusses cases where performance benchmarks may need to be revised downward, if there is a single dominant manufacturer or when evaluating rookie drivers. Further, in Formula 1 technological advantages often dissipate quickly (Rockerbie and Easton 2022), so the returns to additional wind tunnel and CFD investment are likely to remain uncertain.

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Author contributions

CRedit: **John Fry:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Writing – original draft; **Silvio Fanzon:** Conceptualization, Formal analysis, Visualization, Writing – original draft; **Mark Austin:** Conceptualization, Methodology, Writing – review & editing; **Tom Brighton:** Conceptualization, Investigation, Writing – review & editing.

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Appendix: Further performance benchmarking considerations

Revised performance targets with a single dominant manufacturer. Race car performance depends on technological factors such as aerodynamics (Katz 2006), tyre degradation (West and Limebeer 2022), and engine power (Boretti 2020). McLaren's dominance in the 2025 season was attributed to their ability to maintain optimal tyre temperature. This raises the question of how to adjust performance benchmarks in the presence of a clear technological advantage at the manufacturer level. To do this, Equation (A1) should be replaced by

$$\mu_E = E[\textit{Elite driver Finishing Position}] = E[U\{3, 8\}] = 5.5. \quad (\text{A1})$$

Equation (A1) yields the revised simulation results and performance benchmarks in Table A1.

Table A1. Monte Carlo simulation of total season points under the 2025 formula 1 scoring system, with revised parameter $\mu_E = 5.5$.

Category	Mean points	95% CI
Elite driver	195.871	(147–249)
Elite team	391.733	(361–424)

Performance benchmarking for rookie drivers. Suppose a rookie driver joins an elite team but requires one year to adjust before reaching reasonable performance expectations in year two. In this case, performance in the first year would only need to be sufficient to suggest that the targets in Table 3 are likely to be met in the second year. Under this assumption, halving the original performance targets yields an expected points total of 157.73 points, with a 95% confidence interval of 126.5–190.5 points.